

Proof that $\sqrt{5}$ is Irrational

Problem

Prove that $\sqrt{5}$ is irrational.

Solution

Theorem 1. $\sqrt{5}$ is irrational.

Proof. Suppose, for contradiction, that $\sqrt{5}$ is rational. By definition of rationality, there exist integers m, n with $n \neq 0$ such that

$$\sqrt{5} = \frac{m}{n}.$$

Reduction to lowest terms. Let $d = \gcd(m, n)$. Since $n \neq 0$, we have $d \geq 1$. Because $d \mid m$ and $d \mid n$, there exist integers a, b such that $m = da$ and $n = db$. Then $b \neq 0$ and

$$\frac{m}{n} = \frac{da}{db} = \frac{a}{b},$$

We now show $\gcd(a, b) = 1$. Suppose for contradiction that some integer $c > 1$ divides both a and b . Then $c \mid a$ and $c \mid b$, so $c \mid da = m$ and $c \mid db = n$ (since dc divides m and n). More directly: write $a = cq_1$ and $b = cq_2$; then $m = da = dcq_1$ and $n = db = dcq_2$, so $dc \mid m$ and $dc \mid n$. Since $dc > d$ (as $c > 1$ and $d \geq 1$), this contradicts $d = \gcd(m, n)$ being the *greatest* common divisor of m and n . Therefore no integer greater than 1 divides both a and b , i.e. $\gcd(a, b) = 1$.

Thus we may write

$$\sqrt{5} = \frac{a}{b}, \quad a, b \in \mathbb{Z}, \quad b \neq 0, \quad \gcd(a, b) = 1.$$

Deriving divisibility. Squaring both sides gives $5 = a^2/b^2$, so

$$a^2 = 5b^2.$$

Hence $5 \mid a^2$.

We claim $5 \mid a$. To see this, consider the possible residues of a modulo 5. Every integer is congruent to one of 0, 1, 2, 3, 4 modulo 5, and squaring gives

$$0^2 \equiv 0, \quad 1^2 \equiv 1, \quad 2^2 \equiv 4, \quad 3^2 \equiv 4, \quad 4^2 \equiv 1 \pmod{5}.$$

Thus $a^2 \equiv 0 \pmod{5}$ only if $a \equiv 0 \pmod{5}$, i.e. $5 \mid a$.

Write $a = 5k$ for some integer k . Substituting into $a^2 = 5b^2$:

$$(5k)^2 = 5b^2 \implies 25k^2 = 5b^2 \implies b^2 = 5k^2.$$

Therefore $5 \mid b^2$, and by the same residue argument, $5 \mid b$.

Contradiction. We have shown $5 \mid a$ and $5 \mid b$. Since 5 divides both a and b , the integer 5 is a common divisor of a and b . By definition, $\gcd(a, b)$ is the greatest common divisor, so $\gcd(a, b) \geq 5$. This contradicts $\gcd(a, b) = 1$.

Therefore $\sqrt{5}$ is irrational. $\square$